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Paleomagnetic evidence for block rotations in central Hokkaido–south Sakhalin, Northeast Asia

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Abstract

The Eocene to Lower Miocene strata in the Bykov region of south Sakhalin Island, Far East Russia, show a mean paleomagnetic direction of Dec. = 28.7°, Inc. = 55.7°, $\alpha_{95} = 6.7$, and $N = 22$, and a paleomagnetic pole at 66.3°N, 245.7°E, and $A_{95} = 8.8°$ which shows that south Sakhalin rotated $25° \pm 15°$ clockwise with respect to Northeast Asia since the Early Miocene. This pole agrees with that of the Eocene to Middle Miocene sedimentary rocks in central Hokkaido, Japan. It suggests that both south Sakhalin and central Hokkaido have rotated clockwise with respect to Northeast Asia since the Early Miocene. These clockwise rotations can be interpreted as a vertical axis rotation of the blocks distributed, in the north–south direction, more than 1000 km along the dextral Hokkaido–Sakhalin Shear Zone. Considering the development of an en-echelon structure of the basins in the shear zone, we recognized five blocks, each of which had rotated since the Early Miocene. We conclude that this dextral shear is responsible not only for these block rotations but also for the opening of the Kuril Basin and the Japan Sea, and that the shear motion was caused by oblique convergence between the Eurasian and the Okhotsk plates since the Early Miocene. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Miocene; Northeast Asia; paleomagnetic rotation; tectonics; plate boundary

1. Introduction

The boundary between the Eurasian and the North American (Okhotsk) plates runs through Sakhalin Island, Russia, and Hokkaido and northeast Honshu, Japan, as an active N–S trending dextral transgressive shear zone along the eastern margin of the Asian Continent [1,2] (Fig. 1). On-land structural features along this boundary include several N–S trending strike-slip faults cutting through central Hokkaido and Sakhalin, such as the right-lateral Central Hokkaido–Sakhalin Fault, which has been active from the Oligocene to probably recent years as a result of oblique collision between the Eurasian and the Okhotsk plates [3,4]. This dextral shear is also related to the opening of two back-arc basins, the Japan and the Kuril basins [3,5,6]. In central Hokkaido and south Sakhalin, various intensive tectonic events related to the opening tectonics, such as collision-related mountain orogeny [5,7,8]

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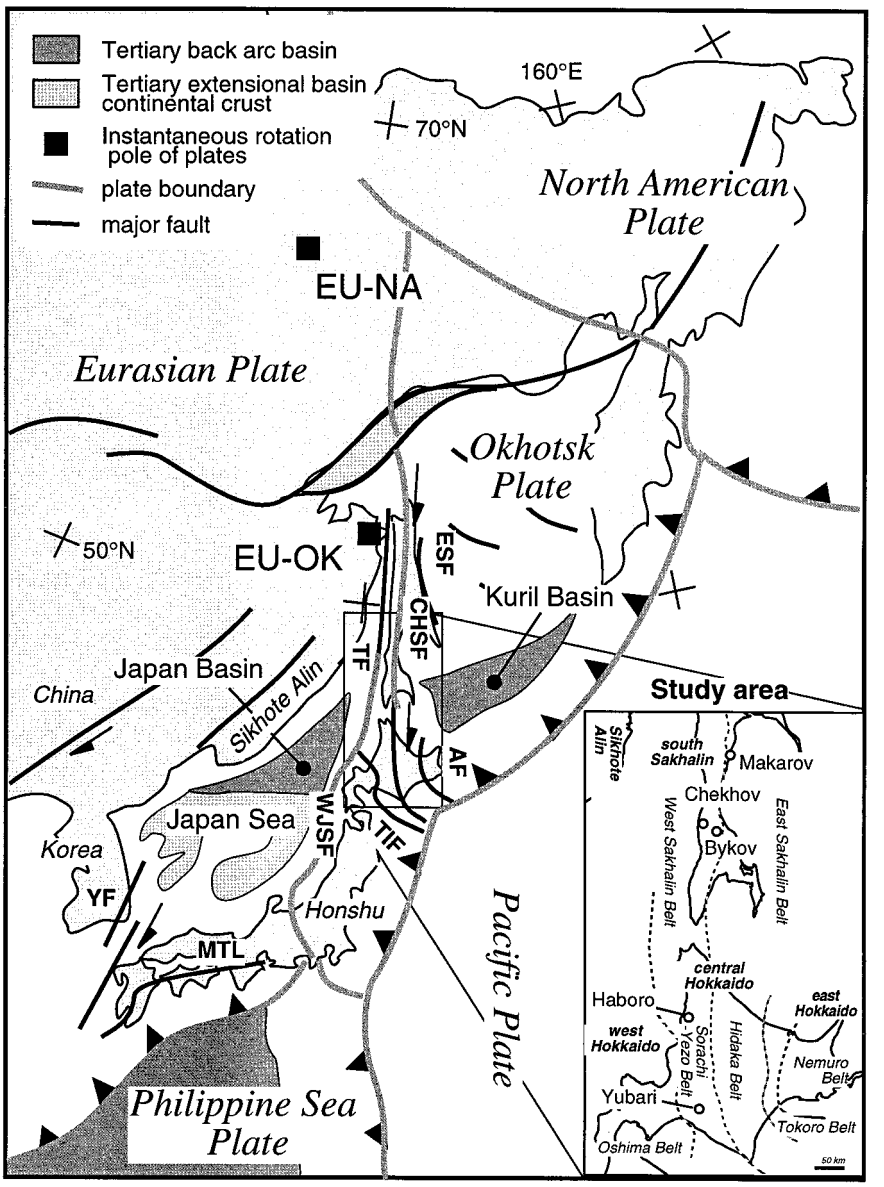


Fig. 1. Simplified present-day tectonic map of Sakhalin and Hokkaido area after [6] and map showing tectonic belts after [3,39] in the study area. The instantaneous rotation poles, *EU-NA* (Eurasian/North American) and *EU-OK* (Eurasian/ Okhotsk), are after [2]. *ESF* = East Sakhalin Fault; *AF* = Abashiri Fault; *CHSF* = Central Hokkaido–Sakhalin Fault; *TF* = Tartary Fault; *TIF* = Tartary–Ishikari Fault; *WJSF* = West Japan Sea Fault; *MTL* = Median Tectonic Line; *YF* = Yangsan and Tsushima Fault.

and episodic volcanism [9,10], also occurred in the Miocene.

To give more quantitative constraints that can serve to explain the Tertiary tectonic history of the Eurasian Plate margin, we carried out a paleomagnetic study on the Tertiary sedimentary strata in south Sakhalin. The new paleomagnetic results, along with previous ones from Hokkaido and northeast Honshu, and re-examination of on-land geologic features enabled us to provide a new model that better explains the tectonic development of convergent plate margins.

2. Geological outline and sampling

The basement of Sakhalin is composed of Cretaceous to lower Paleogene accretionary complexes accumulated along the N–S trending, westward paleo-subduction zone between the Eurasian Plate and the three oceanic plates, the Izanagi, the Kula and the Pacific plates [11]. Sakhalin is structurally divided into two parts, the Eastern Sakhalin Belt and the Western Sakhalin Belt, by the N–S trending Central Hokkaido–Sakhalin Fault (Fig. 1). In the Western Sakhalin Belt, thick fore-arc basin deposits of the post-Early Cretaceous are widely distributed, dipping westward with N–S trending strike. They are overlain by Cenozoic clastic and volcanic deposits, which are characterized by NW–SE trending en-echelon faults and folds [3,4]. This belt is equivalent to the Sorachi–Yezo Belt of central Hokkaido (Fig. 1). In contrast, the Eastern Sakhalin Belt, the northern extension of the Hidaka Belt of central Hokkaido (Fig. 1), is characterized by many N–S trending strike-slip faults, and the basement consists of Cretaceous accretionary complexes overlain by Neogene deposits and is intruded by igneous rock bodies.

The Tertiary System in the Western Sakhalin Belt is lithostratigraphically divided into ten formations, from the Lower Duye Formation at the bottom to the Maruyama Formation at the top [12,13]. The stratigraphic summary is illustrated in Fig. 2. The Lower Duye Formation unconformably overlies the Late Cretaceous Krasnoyarka Formation and consists of coal-bearing terrigenous sandstones, correlated

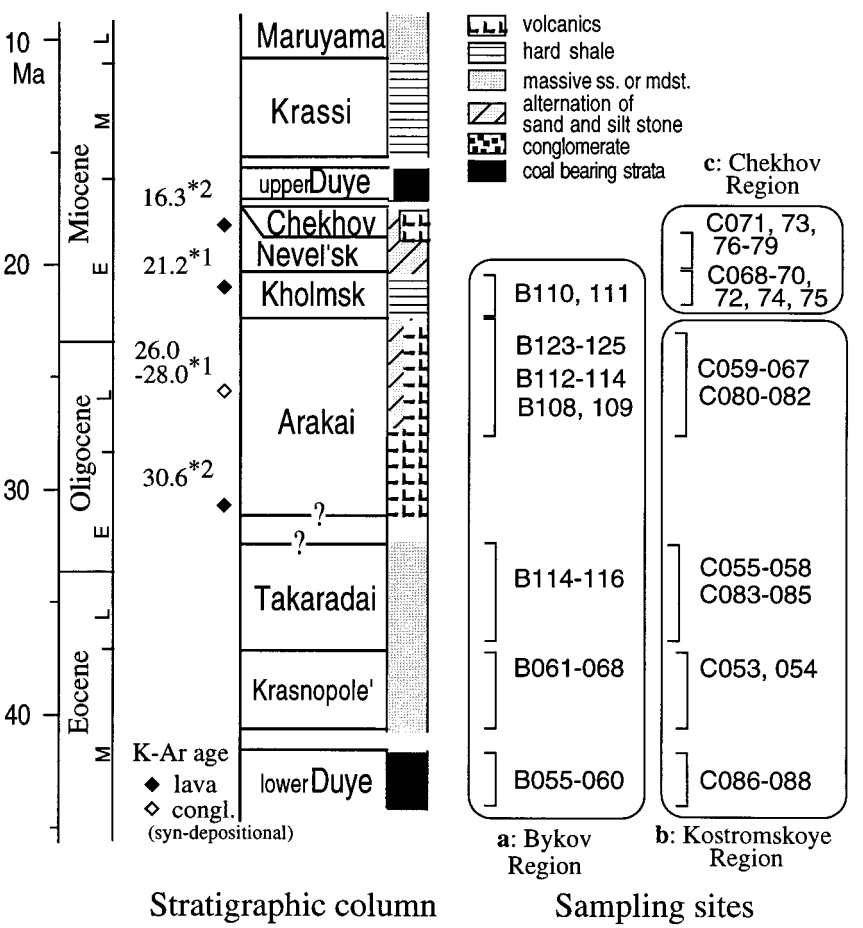


Fig. 2. Stratigraphic column of the Tertiary systems in south Sakhalin after [10], with horizons of sampled sites and K–Ar ages after [10] (*1) and [15] (*2).

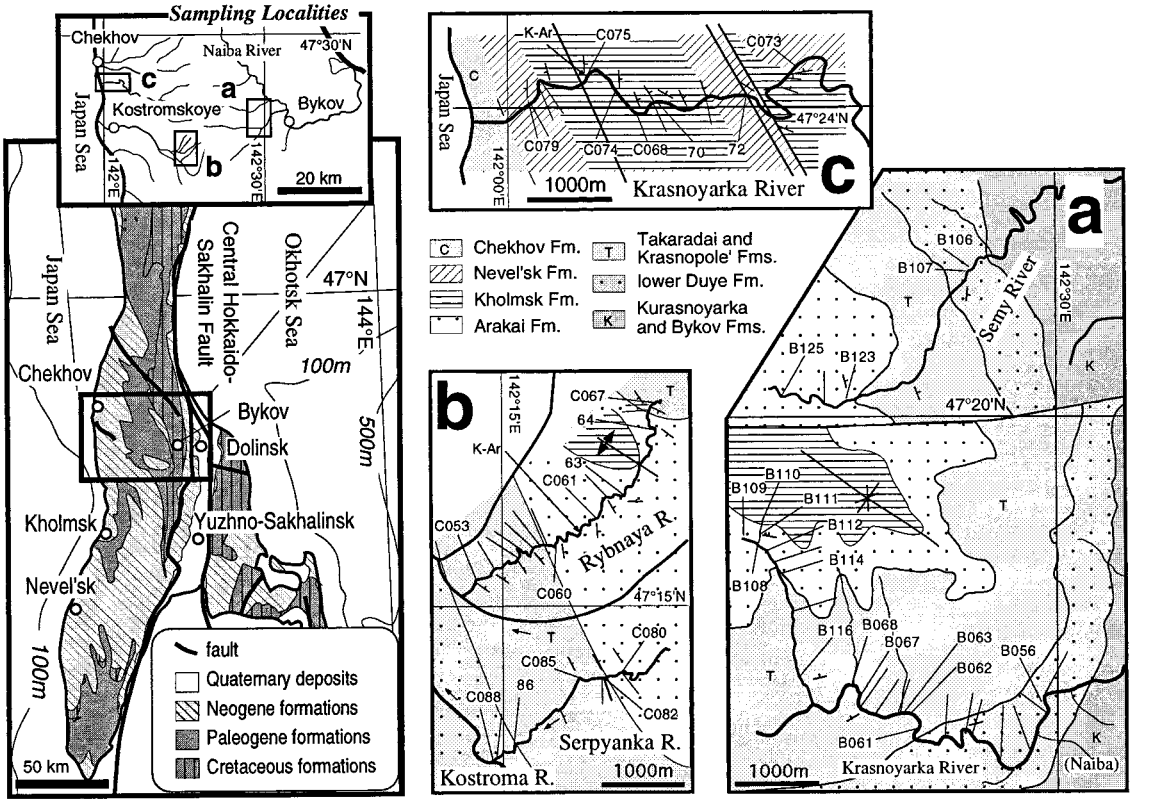


Fig. 3. Geological map of study area and localities of the sampled sites pointed out by thin straight lines in the Buikov region (a), the Kostromskoye region (b), and the Chekhov region (c). Geological map is after [40].

with the Paleogene coal-bearing formations in central Hokkaido. The Krasnopole' and the Takaradai formations yield foraminifera [13,14], which also occur in the Poronai and the Momijiyama formations of the Lower Eocene to the Upper Oligocene in central Hokkaido. The Arakai Formation consists of thick volcanic sandstone and conglomerate derived from intensive andesitic volcanism. The andesitic boulders in the upper part have a K–Ar age of 28–26 Ma [10]. The Kholmsk Formation is characterized by siliceous hard shales. An andesitic lava flow intercalated in this formation has a K–Ar age of 21 Ma [10]. Thick volcaniclastic layers of the Chekhov Formation indicate episodic basic volcanism at about 17 Ma [15]. The Upper Duye Formation contains warm-climate flora [16] similar to that in the Takinoue and the Chikubetsu formations in central Hokkaido. The uppermost Maruyama Formation is composed of sandstone and diatomaceous shale with its upper part containing *Fortipecten takahashii* [17], an index molluscan species of the Pliocene.

Samples for paleomagnetic study were collected from 29 sites along the Krasnoyarka River and the Semy River near Bykov Town (Fig. 3a). Samples were also collected from 24 sites along the upstream of the Kostroma River, east of Kostromskoye Town (Fig. 3b), and 18 sites from the Krasnoyarka River near Chekhov City (Fig. 3c). Stratigraphic horizons of the sampling sites are summarized in Fig. 2. At each site in the river beds, more than six core samples were taken from a single to a maximum of three beds, less than 5 m in thickness using a magnetic compass for orientation and a gas-powered core drill. One or two cylindrical specimens of 24 mm in diameter and a height of 22 mm were prepared from each core for measurement.

3. Paleomagnetism

3.1. Measurements

All specimens were measured with a 2G Enterprises 760 three-axis cryogenic magnetometer, after progressive alternating field (AF) or thermal demagnetization in a magnetically shielded laboratory at Kochi University. At first, two or three pilot specimens were taken from each site and subjected to more than ten-step progressive AF and thermal demagnetization. AF demagnetization was done from 2.5 to 10 mT with 2.5 mT increments and up to 45 mT with 5 mT increments, additionally up to 70 mT with 10 mT intervals for some specimens. Supplementary thermal demagnetization with steps above 200ºC was applied to those specimens for which AF demagnetizations were judged insufficient. Normally, thermal demagnetization was done in steps from 120ºC with 40ºC increments until 600ºC, or until the intensity became too weak to measure. Low-field bulk magnetic susceptibility was measured each time after a cooling run to monitor alteration of magnetic minerals. Thermal demagnetization was also terminated when random changes in remanence direction or rapid increase in magnetic susceptibility were encountered. More than seven-step progressive demagnetization by AF or thermal demagnetization were used on the remaining specimens after the pilot studies.

Characteristic remanent magnetization (ChRM) components were isolated by applying the method of Kirschvink [18] to vector segments composed of at least four data points with a maximum angular deviation (MAD) smaller than 15. When demagnetization great circles were available, the indirect method of McFadden and McElhinny [19] using demagnetization plane data directing to suspected magnetic component and at least two demagnetization line data was applied to estimate a site mean direction and associated Fisher statistics. Each site mean direction was calculated from more than six data of individual specimens from at least four cores.

3.2. Results

ChRM components were isolated from about 240 specimens of a total of 630 measured specimens from 71 sites. The ChRM components of other specimens could not be isolated because either their MAD value became larger than 15º or their stable vector end-points were less than three. The demagnetization great circles can be isolated from about 170 of the specimens.

Initial NRM intensities of coarse to fine sandstone from the Lower Duye Formation generally ranged from 2×10^{-3} to 2×10^{-4} A/m. ChRMs of half of the specimens from site B056 were isolated above 160ºC or 10 mT, and those of site C088 above 240–400ºC or 30 mT (Fig. 4a). Site mean directions of the two sites were calculated by the indirect method of McFadden and McElhinny [19] using remagnetization circles isolated below 20–30 mT or 200–400ºC (Fig. 5a). Initial NRM intensities of massive fine sandstone and siltstone from the Krasnopole' and the Takaradai formations ranged from 2×10^{-3} to 2×10^{-4} A/m, except for three sites (B066, B067, and B068) that had a 10^{-2} A/m intensity. About one-fifth of the specimens from the Krasnopole' and the Takaradai formations possess ChRMs above 10–20 mT and 160–240ºC, but they showed erratic behavior above 400ºC associated with an increase in magnetic susceptibility (Fig. 4b). Site mean directions of B062, B063, B116 and C085 were estimated by the indirect method (Fig. 5b). Mean directions of the other 14 sites from the Krasnopole' and the Takaradai formations and nine sites from the Lower Duye Formation could not be calculated due to a lack of direct and indirect data.

Site mean directions of the Eocene to the Early Oligocene were determined from eight sites of the Lower Duye, the Krasnopole', and the Takaradai formations (Table 1), showing an antipodal relationship with a NNE–SSW trend after tilt correction (Fig. 6). A considerable increase in the precision parameter k after tilt corrections of the overall mean directions from 13.7 to 40.0 indicates that the magnetizations were acquired before the tilting occurred. The reversal test by McFadden and McElhinny [20] was positive at the 95% confidence level; the observed angle between the normal and the reversed directions $\gamma_o = 13.6^o$ does not exceed the simulated angle $\gamma_c = 20.5^o$. The overall mean of the tilt-corrected directions is Dec. $= 16.7^o$, Inc. $= 55.6^o$, $\alpha_{95} = 8.9^o$, $k = 40.0$, and $N = 8$ with a paleomagnetic pole at 73.7ºN, 267.2ºE with $A_{95} = 11.4^o$ (Table 1).

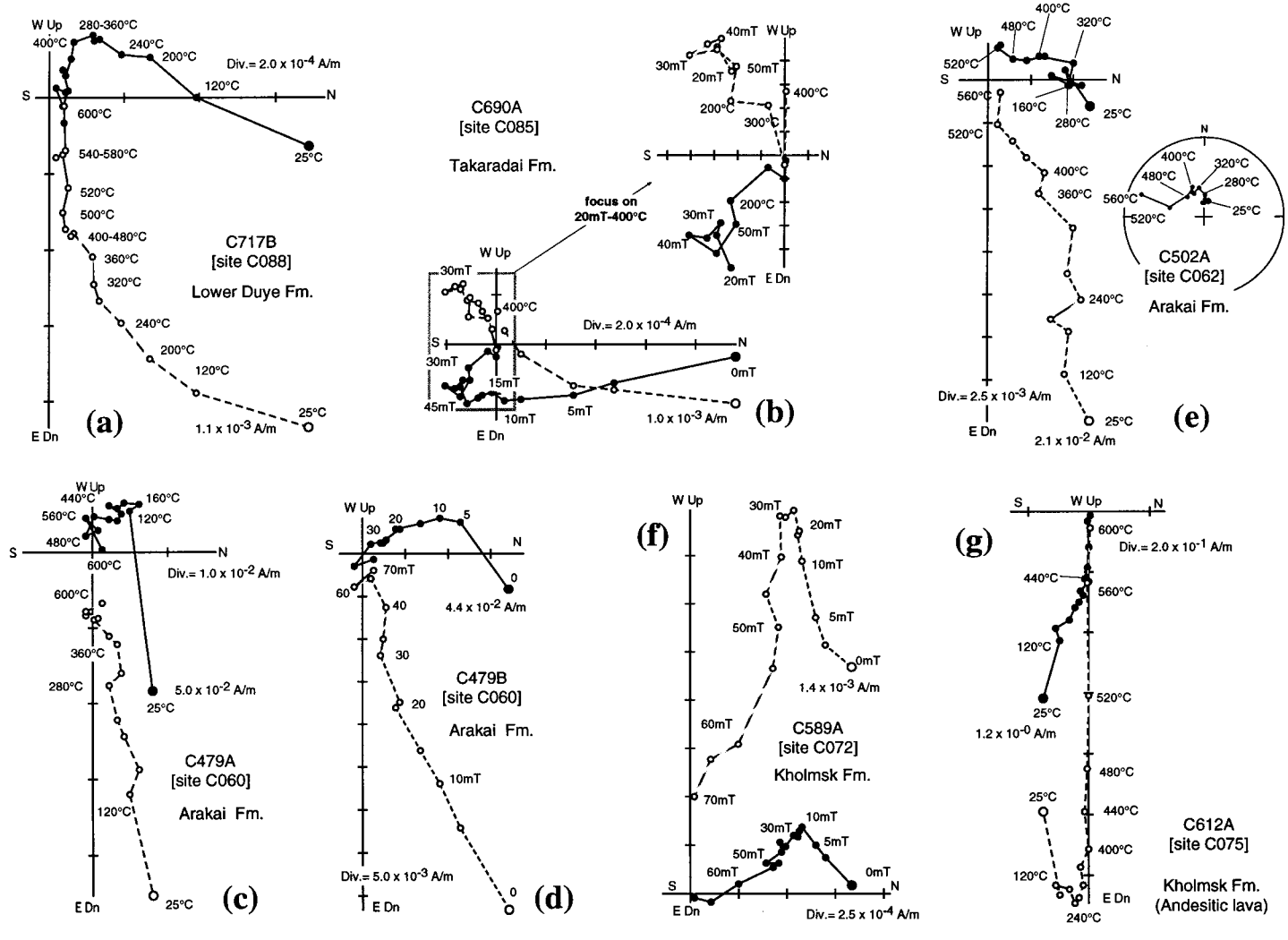


Fig. 4. Examples of progressive demagnetization result on the orthogonal end-point diagram in in-situ coordinates. Solid (open) circles are projections onto the horizontal (N–S vertical) plane. Demagnetization end-points are also on the lower hemisphere of the equal-area projection in (e).

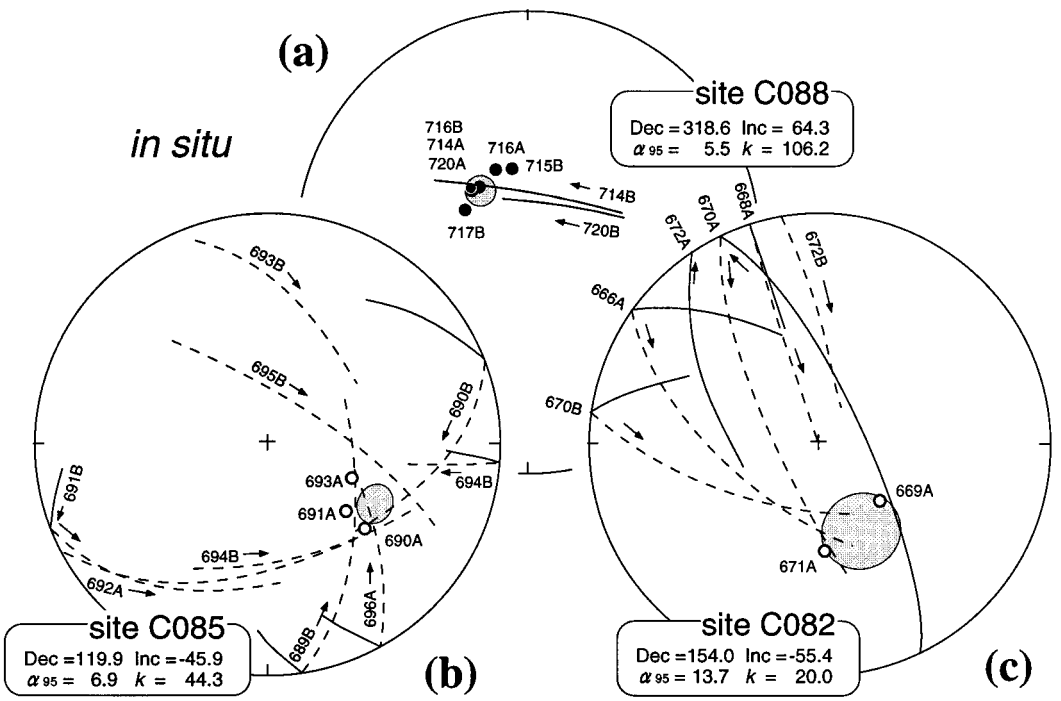


Fig. 5. Examples of the analysis for computing site mean directions using the method of [19]. Solid circles and lines are on the lower hemisphere of the equal-area projection, and open circles and dashed lines are on the upper hemisphere. Circles are plots of magnetization directions isolated by direct line fit. Great circles are the best-fit demagnetization trajectories which directions are indicated by small arrows. Shaded ovals are the 95% confidence limits of site mean directions.

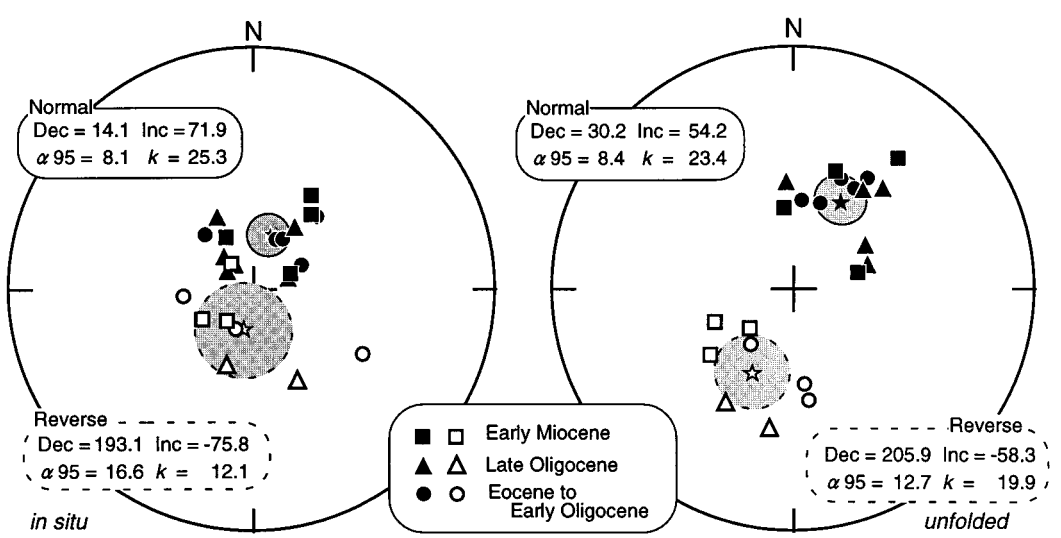


Fig. 6. Site mean directions and overall mean directions in geographic (left) and stratigraphic (right) coordinates on the equal-area projections. Solid (open) symbols are on the lower (upper) hemisphere. Ovals indicate 95% confidence limit of the overall mean directions of normal or reversed polarity. Numeric data are given in Table 1.

Table 1
Summary of paleomagnetic results for the Tertiary formations in south Sakhalin

Core site	Fm.	Location		Bedding		In situ		Unfolded		Statistics			Pole	
		lat. (°N)	long. (°E)	strike (°)	dip (°)	dec. (°)	inc. (°)	dec. (°)	inc. (°)	α_{95} (°)	k	N	lat. (°N)	long. (°E)
Eocene to Early Oligocene														
B056	LD	47.30	142.5	59	−22	205.7	−75.1	173.1	−57.3	10.2	26.9	9 (2)	79.3	353.3
C088	LD	47.23	142.3	−27	42	318.6	64.3	24.3	47.9	5.5	106.2	8 (2)	63.8	267.6
B062	KR	47.30	142.5	68	−23	63.5	71.6	17.3	59.5	11.1	22.8	9 (1)	75.7	255.5
B063	KR	47.30	142.5	59	−20	263.4	−66.1	216.5	−66.1	18.8	16.6	6 (4)	65.7	216.2
B067	KR	47.30	142.5	63	−14	26.5	70.6	5.8	60.0	7.4	83.1	6	82.4	287.4
B114	TK	47.32	142.5	−61	20	31.3	69.3	30.7	49.3	8.0	70.7	6	60.9	257.5
B116	TK	47.31	142.5	−81	15	42.0	56.5	33.6	43.2	14.9	17.6	7 (1)	55.5	260.5
C085	TK	47.24	142.3	−28	42	119.9	−45.9	171.3	−51.3	6.9	44.3	12 (9)	73.4	349.0
Mean				in situ		16.1	73.0			15.5	13.7	8	73.7	267.2
				unfolded				16.7	55.6	8.9	40.0		A_{95} = 11 .4	
Late Oligocene														
B108	AR	47.32	142.5	−51	16	33.4	63.8	35.7	47.9	6.9	65.5	8	56.9	253.3
B109	AR	47.32	142.5	−41	20	200.1	−62.6	211.4	−44.2	8.6	41.1	9 (6)	57.3	262.2
B112	AR	47.32	142.5	−51	16	334.6	62.1	355.4	52.7	6.0	86.1	8	75.6	338.1
C060	AR	47.26	142.3	−8	36	318.9	74.8	58.3	60.4	3.9	239.4	7	48.9	218.4
C061	AR	47.26	142.3	−4	34	310.3	79.0	71.4	63.2	9.2	44.4	7	42.1	207.6
C080	AR	47.24	142.3	−34	47	324.4	79.5	41.7	42.4	15.5	19.7	6	50.0	252.0
C082	AR	47.24	142.3	−31	34	154.0	−55.4	190.5	−40.9	13.7	20.0	8 (6)	64.8	299.2
Mean				in situ		347.9	70.7			11.0	31.0	7	61.5	248.1
				unfolded				31.8	52.2	13.1	22.0		A_{95} = 17.5	
Early Miocene														
B110	KH	47.32	142.5	−35	23	31.3	51.7	38.6	30.2	11.5	28.6	7	45.6	263.6
B111	KH	47.32	142.5	−61	12	333.0	70.0	352.8	61.6	5.7	95.3	8	83.2	13.4
C068	KH	47.40	142.0	−22	6	220.7	−75.6	228.5	−70.1	6.1	47.4	14 (11)	59.2	201.2
C072	KH	47.40	142.0	−58	39	321.3	−77.8	231.7	−53.5	12.9	30.0	6 (2)	49.4	231.9
C074	KH	47.40	142.0	−4	10	239.2	−69.8	247.6	−60.6	9.4	24.6	12 (10)	43.0	212.9
C075	KH	47.40	142.0	−4	10	70.0	77.0	76.8	67.2	3.4	161.6	12	41.5	198.8
C073	NV	47.41	142.1	69	−18	37.6	57.7	19.9	45.9	8.6	50.2	7	64.6	277.2
Mean				in situ		40.2	72.7			13.0	22.6	7	60.0	229.0
				unfolded				40.3	58.1	14.4	15.7		A_{95} = 19.4	
Overall mean				in situ		13.8	73.3			7.3	19.0	2 (2)	66.3	245.7
				unfolded				28.7	55.7	6.7	22.7		A_{95} = 8.8	

Fm. = formation; LD = Lower Duye Fm.; KR = Krasnopole' Fm.; TK = Takaradai Fm.; AR = Arakai Fm.; KH = Kholmsk Fm.; NV = Nevel'sk Fm.; lat./long. (location) = latitude/longitude; bedding = dip clockwise from strike; dec./inc. = declination/inclination; α_{95} = the radius of 95% confidence cone; k = Fisherian precision parameter; N = number of specimens used for calculation of the means, number of demagnetization great circIes are in parentheses if the method of [22] was used; lat./long. (pole) = north latitude/east longitude of paleomagnetic pole; A_{95} = 95% confidence limit for mean paleomagnetic pole.

The NRM intensities of the Arakai Formation vary considerably according to lithology, the volcanic sandstone being of the 10^{-2} A/m order and the fine sandstone ranging from 5×10^{-3} to 5×10^{-4} A/m. The more strongly magnetized specimens showed a more stable behavior for demagnetizations. Specimens from five out of nine sites of volcanic sandstone (C060, C061, C080, B108, and B112) revealed ChRMs above 160–360°C (Fig. 4c) and 20 mT (Fig. 4d). Although some specimens of sites C062–C067 seem to have a single magnetic component with in-situ directions close to the recent geo-

magnetic field direction, vector end-points of several specimens of each site moved even in reversed direction along the great circle (Fig. 4e). We rejected these six sites because of overprinting of the recent geomagnetism. Many of the weakly magnetized sandstone specimens showed unstable behavior, so that site mean directions were determined for only two out of ten sites (C082 and B109), whose mean directions were calculated with remagnetization circles (Fig. 5c).

A tilt-corrected, overall mean direction of the Late Oligocene including the Arakai Formation is Dec. = 31.8°, Inc. = 52.2°, α_{95} = 13.1°, k = 22.0, N = 7 with a paleomagnetic pole at 61.5°N, 248.1°E with A_{95} = 17.5°, which was determined from the seven site mean directions with mixed magnetic polarities. A fold test by McFadden [21] was applied but no statistical significance was concluded because the parameter ξ_2 in both the in-situ position (0.218) and the unfolded position (2.512) did not exceed the 95% confidence limit of 3.086. This is probably because all the sampled sites are located in a single westward-dipping fold limb. Although the fold test was not effective, the positive result of the reversal test [20] with the observed angle γ_o = 16.8° being smaller than γ_c = 23.6° and coherence of directions implies that the magnetizations are primary.

Specimens from the Kholmsk and the Nevel'sk formations were taken from diatomaceous shale and volcanic fine sandstone, except those from site C075 taken from andesitic lava flows. Their NRM intensities ranged from 5×10^{-3} to 2×10^{-4} A/m. ChRMs appeared above 40–60 mT and 200–300°C, but became so unstable above 50–70 mT and 400°C that it was hard to isolate stable components without using remagnetization circles (Fig. 4f). Stable magnetic components could not be isolated from more than half of the specimens. Mean directions of sites C075, C073, B110, and B111 were calculated using individual ChRM directions, while those of sites C068, C072, and C074 were estimated by the indirect method. The specimens from one andesitic lava flow at site C075 were exceptional in that the NRM are of the 1.0 A/m order and considerably stable up to 600°C. The ChRMs appeared above 30 mT or 480°C (Fig. 4g).

An overall mean direction after tilt correction of the Early Miocene including the Kholmsk and the Nevel'sk formations was determined from the seven sites to be Dec. = 40.3°, Inc. = 58.1°, α_{95} = 14.4°, k = 15.7, and N = 7 with a paleomagnetic pole at 60.0°N, 229.0°E with A_{95} = 19.4° (Table 1). No better grouping of statistical significance was achieved by tilt correction. The parameter k decreased from 22.6 to 15.7 by 100% tilt correction, but it showed a slight increase (23.4) by a 21% partial unfolding. However, no statistical conclusion could be drawn from the McFadden fold test [21] resulting in ξ_2 in the in-situ position (1.760) and in the totally unfolded position (1.074), both being smaller than the 95% confidence limit of 3.086. In contrast, the reversal test was positive, with γ_o = 15.8° smaller than γ_c = 27.6°, which, lacking a statistically meaningful fold test, we regarded as evidence for paleomagnetic stability.

3.3. Summary

A total of 22 site mean directions were acquired from the Eocene to Early Miocene age units in south Sakhalin Island. Because there is no statistical difference in the mean directions with age and with region taking consideration of their 95% confidence error, we averaged all the 22 site mean directions to determine an overall mean direction.

The overall mean direction was calculated, after converting the reversed directions to normal, to be Dec. = 13.8°, Inc. = 73.3°, N = 22 with α_{95} = 7.3° in geographic coordinates, and Dec. = 28.7°, Inc. = 55.7° with α_{95} = 6.7° in stratigraphic coordinates (Fig. 6). Slightly better grouping was achieved upon applying tilt corrections as observed in k increased from 19.0 to 22.7. An antipodal relationship holds for all the 22 directions as evidenced by γ_o = 4.7° not exceeding γ_c = 15.1°.

The mean paleomagnetic pole of south Sakhalin is 66.3°N, 245.7°E with A_{95} = 8.8°, which agrees with statistical uncertainties of paleomagnetic data from central Hokkaido, specifically from the Haboro and Yubari regions, including one of Oligocene to early-Middle Miocene age, 69.5°N, 210.2°E with A_{95} = 14.4° and N = 11, and one of middle-Middle Miocene age, 70.0°N, 255.5°E with A_{95} = 12.3° and N = 12 [22] (Fig. 7; Table 2). This result implies that south Sakhalin Island and central Hokkaido can be regarded as a single tectonic zone that rotated

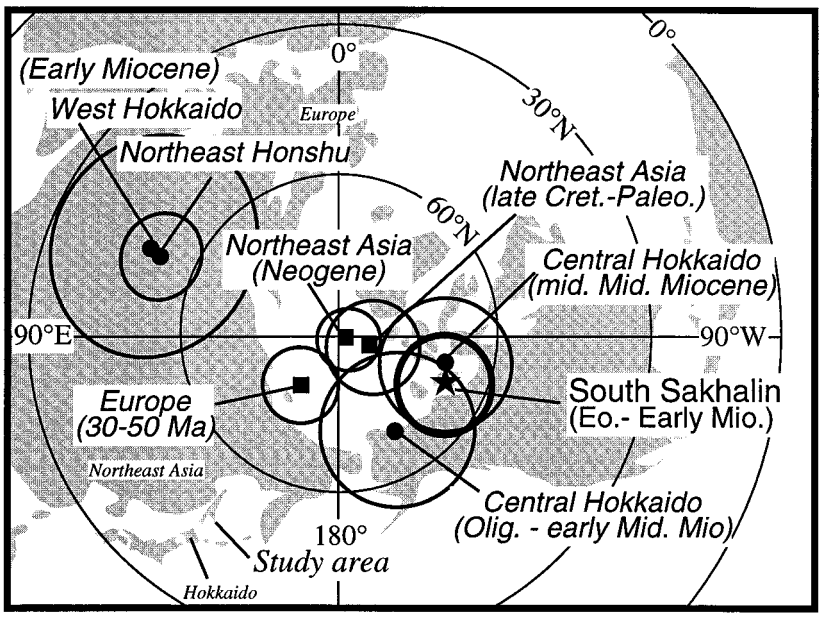


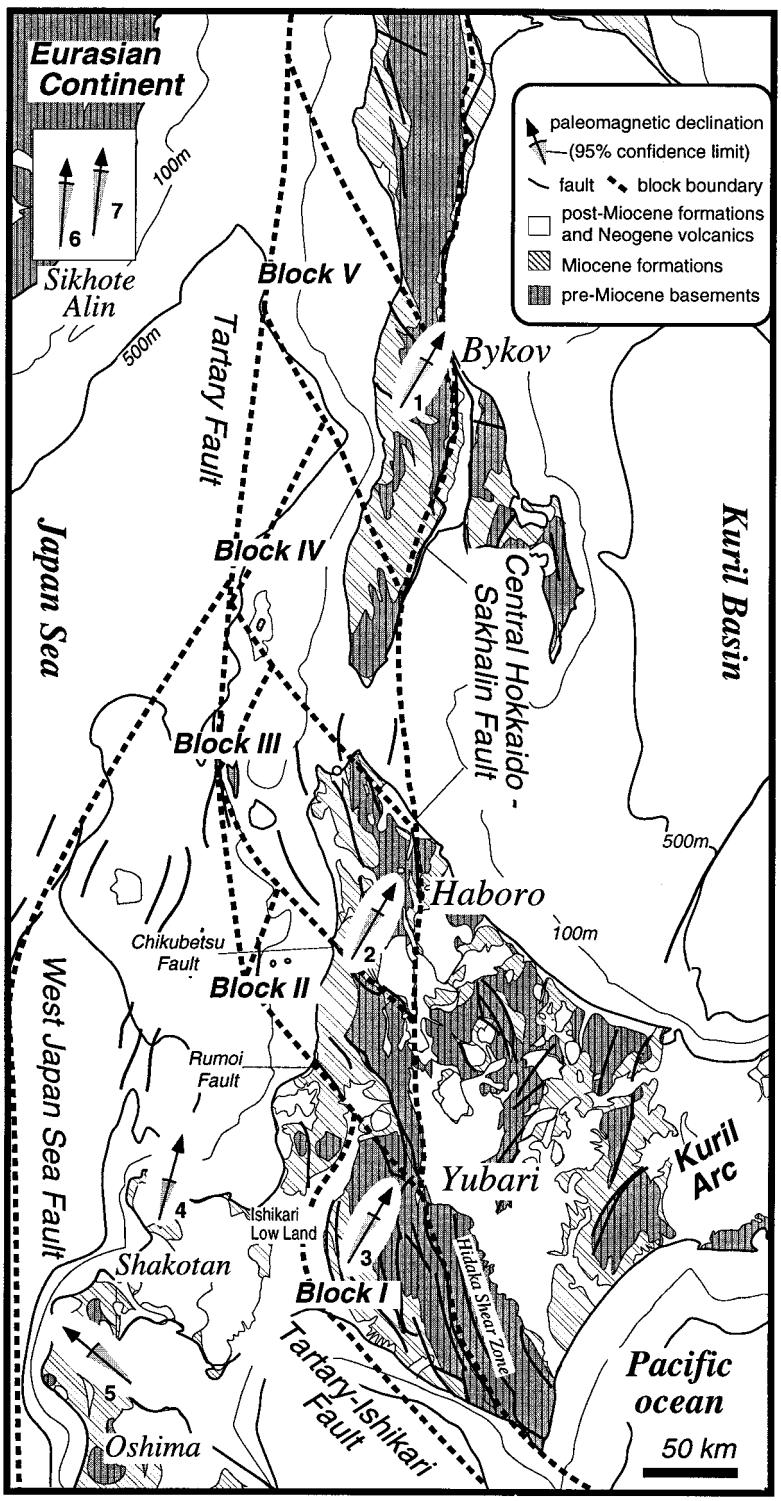
Fig. 7. Paleomagnetic pole positions and associated circles of 95% confidence for south Sakhalin and reference regions. Numeric data and associated statistics of each reference paleomagnetic pole are given in Table 2.

Table 2
Paleomagnetic pole positions of various areas in Eurasia and relative rotations of south Sakhalin with respect to them

Area	Age	Lat. (°N)	Long. (°E)	A_{95} (°)	R	ΔR	F	ΔF	Reference
South Sakhalin	Eocene–Early Miocene	66.3	245.7	8.8	–	–	–	–	this study
Central Hokkaido	Olig.–early-Middle Miocene	69.5	210.2	14.4	−1.5	20.5	−11.0	10.5	[22]
Central Hokkaido	middle-Middle Miocene	70.0	255.5	12.3	6.4	15.1	0.7	11.5	[22]
Northeast Asia	Late Cretaceous–early Paleogene	84.1	250.5	8.8	21.6	13.4	−6.6	8.7	[23]
Northeast Asia	Neogene	88.3	247.0	6.4	27.2	11.3	−7.9	7.6	[a]
Sikhote Alin		86.8	257.6	2.4	25.4	9.3	−7.1	6.8	[24]
Korea		86.6	349.4	8.9	31.7	13.2	−5.9	8.7	[25]
North China Block		85.2	192.5	5.1	23.8	10.5	−10.4	7.1	[26]
Europe	30–50 Ma	78.0	140.0	7.0	30.5	13.1	−16.5	7.3	[27]
West Hokkaido	Early Miocene	50.4	64.7	19.9	82.6	22.3	−1.8	15.5	[29]
Northeast Honshu	Early Miocene	53.3	66.3	7.8	77.5	12.0	−1.7	8.7	[29]

Lat., Long. = latitude and east longitude of paleomagnetic pole; A_{95} = the radius of 95% confidence circle probability; R and F = relative rotation and flattening indices of [43] with errors of 95% confidence level (ΔR and ΔF) corrected using factors of [44].
[a] Paleomagnetic pole of Neogene Northeast Asia was calculated following that of Sikhote Alin, Korea, and North China Block.

Fig. 8. Paleomagnetic declinations with 95% confidence limit and possible block boundaries of five rotational blocks in the Hokkaido–Sakhalin Shear Zone. *1* = south Sakhalin (Eocene–Middle Miocene) [this study]; *2* = Haboro region (Eocene–Middle Miocene) [22]; *3* = Yubari region (Oligocene–Middle Miocene) [22]; *4* = Shakotan Peninsula (Early–Late Miocene) [28]; *5* = Oshima Peninsula (Early Miocene) [29]; *6* = Sikhote Alin (Neogene) [24]; *7* = Sikhote Alin (Late Cretaceous–Early Eocene) [23]. Geological map offshore fault map is after [40–42].


Eurasian Continent
paleomagnetic declination
(95% confidence limit)
fault
block boundary
post-Miocene formations and Neogene volcanics
Miocene formations
pre-Miocene basements
100m
500m
Sikhote Alin
Block V
Tartary Fault
Bykov
Japan Sea
Block IV
Central Hokkaido - Sakhalin Fault
Kuril Basin
Block III
500m
Haboro
100m
Chikubetsu Fault
Block II
West Japan Sea Fault
Rumoi Fault
Kuril Arc
Yubari
Ishikari Low Land
Shakotan
Hidaka Shear Zone
Block I
Tartary-Ishikari Fault
Pacific ocean
50 km
Oshima

clockwise after the Early Miocene. We regarded the relative rotation of south Sakhalin during the Eocene to Early Miocene as a $25^{o} \pm 15^{o}$ clockwise one with respect to Northeast Asia, as relative rotations of south Sakhalin with respect to Northeast Asia led to $21.6^{o} \pm 13.4^{o}$ and $27.2^{o} \pm 11.3^{o}$, during the Late Cretaceous to Early Eocene (Sikhote Alin, Russia [23]) and during the Neogene (averaged from Sikhote Alin [24], Korea [25], and the North China Block [26]), respectively (Table 2). Relative rotation of south Sakhalin with respect to Europe [27] is similarly regarded as a $30.5^{o} \pm 13.1^{o}$ clockwise one. The south Sakhalin and central Hokkaido region contrast with west Hokkaido and northeast Honshu that experienced counterclockwise rotation with respect to East Asia in the middle-Middle Miocene [28,29].

4. Discussion

NW–SE to NNW–SSE trending en-echelon faults and folds are commonly observed in the Western Sakhalin and Sorachi-Yezo Belts. These en-echelon faults are bounded by the parallel N–S trending Central Hokkaido–Sakhalin and Tartary-Ishikari Faults (Fig. 8). Kimura et al. [4] suggested that this fault-bounded zone has been a dextral shear zone since the Early Miocene and the Central Hokkaido–Sakhalin Fault has been active as a right-lateral strike-slip fault from the Miocene to the Pliocene. Fournier et al. [3] also indicated that the Tym-Poronai Fault, a northern part of the Central Hokkaido–Sakhalin Fault, has moved right-laterally associated with en-echelon folds from the Miocene to the Present. Jolivet and Huchon [30] concluded that the Hidaka Shear Zone, which is the southern part of the Central Hokkaido–Sakhalin Fault, has been active with a right-lateral strike-slip component from the Oligocene to the end of the Middle Miocene. The Chikubetsu and the Rumoi Faults, typical NW–SE trending faults connecting the parallel N–S boundary faults, have been interpreted to be active since the Early Miocene [31]. The Chikubetsu Fault has also moved left laterally with a displacement of at least 40 km since the Late Miocene [32].

Paleomagnetic directions of the fault-bounded belt in both south Sakhalin (this study) and central Hokkaido [22] show about 30º clockwise rotation with respect to Northeast Asia since the Middle Miocene. These synchronous rotations cannot be explained as a ridged coherent block, including Sakhalin and Hokkaido, drifting from Asia undergoing clockwise rotation due to geographical limitation, but the rotations are in good accordance with a regional dextral transpressive shear model. We call hereafter this fault-bounded dextral shear zone the Hokkaido–Sakhalin Shear Zone, and interpret the clockwise rotations as block rotation in a dextral shear zone in a similar manner to the model of Ron et al. [33]. Paleomagnetic data suggest that this shear zone has been active as a dextral shear zone along the convergence boundary between the Eurasian and the Okhotsk plates after the opening of the Japan Sea ceased [6,22,29].

On the basis of sequential geologic features, we recognized five individual blocks in the Hokkaido–Sakhalin Shear Zone (Fig. 8). Each block is assumed to be bounded by N–S trending faults as boundary fault and subdivided by NW–SE trending faults as rotational fault. Each angle between rotational and boundary faults is about 45º. The Hokkaido–Sakhalin Shear Zone exhibits a basin and range type structure (Fig. 8), which started to develop since the early-Middle Miocene [34,35]. The synchronism of the basin formation and the clockwise rotation suggests a tectonic model, in which the oblique collision of the proto-Kuril Arc to central Hokkaido and Sakhalin had caused dextral shear rotation and mountain building (Fig. 9).

In this model, the proto-Kuril Arc moved southward from the Early to Late Miocene, giving rise to the opening of the Kuril Basin [10]. This proto-Kuril Arc then collided against central Hokkaido and caused rapid accumulation of the Kotambetsu and the Kawabata formations [34,35]. Ages of these formations suggest that the collision-related uplift has migrated from north to south through the Miocene [36]. The southward drift of the arc is consistent with the southward shift of vigorous volcanism in central and eastern Hokkaido from 15 to 9 Ma [37,38]. The episodic volcanism dated to be 23–21 Ma along the Central Hokkaido–Sakhalin Fault around Makarov City, eastern coast of south Sakhalin [10], may represent the earliest stage of the shift of volcanism. We further postulate that the spreading of the Kuril Basin took place in a synchronous manner with the

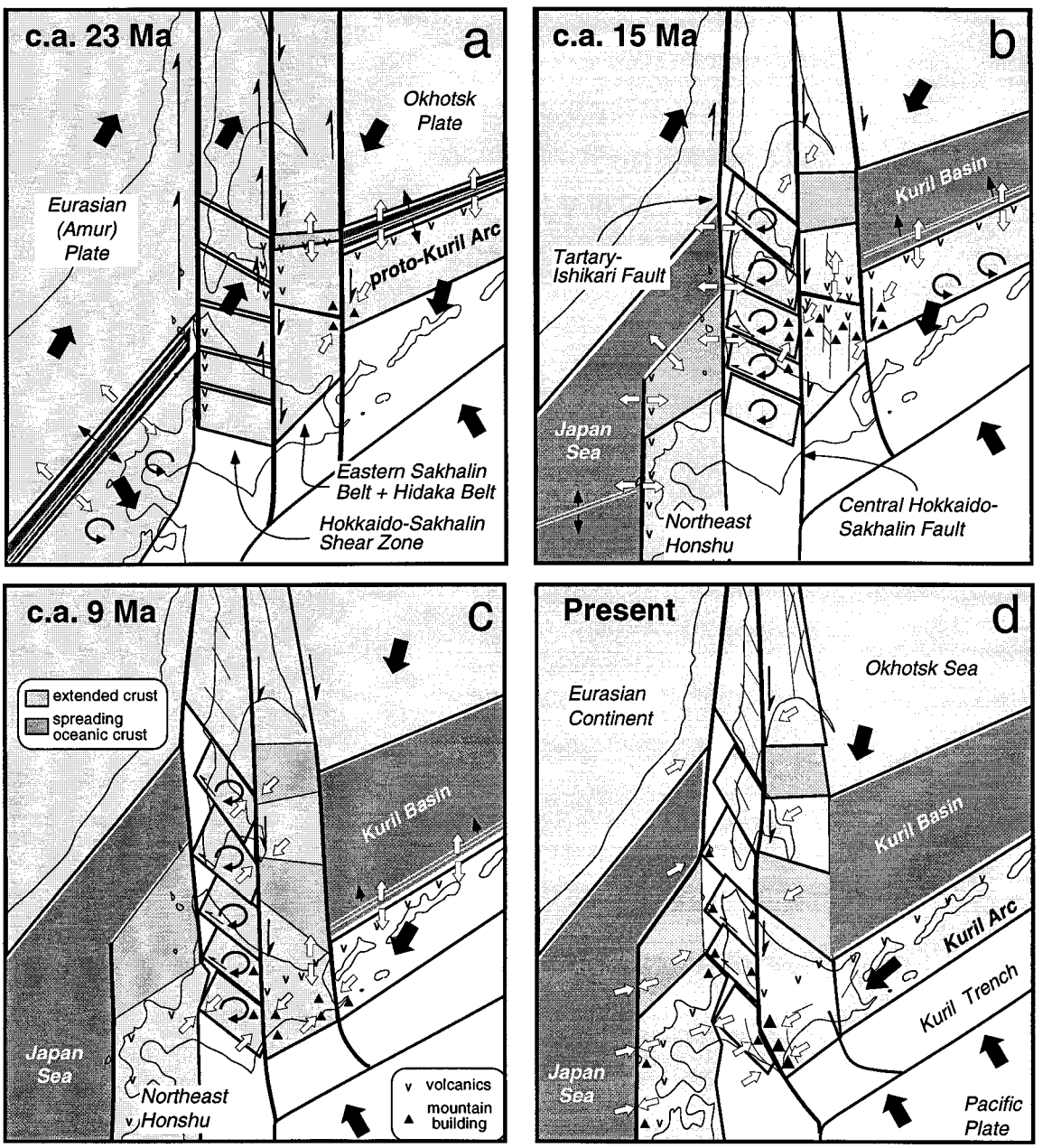


Fig. 9. Tectonic model of Sakhalin and Hokkaido since the Early Miocene. Tectonic framework of this model is after [6]. Solid (open) arrows represent direction of movement of plates and blocks (main stress field directions). Thick lines are block boundary faults and thin lines in the blocks are minor faults. Thin arrows along faults represent direction of faults. At the beginning of the Miocene (a), dextral shear was dominant in the Hokkaido–Sakhalin Shear Zone as a result of oblique collision of the Eurasia and the Okhotsk plates [3,5]. From the Early to middle-Middle Miocene (a–b), southward drift of the proto-Kuril Arc caused dextral transpressive shear along the Central Sakhalin–Hokkaido Fault and the segmented blocks started clockwise rotation. E–W extension took place along the Tartary-Ishikari Fault, which could be related to the opening of the Japan Sea. Crustal uplift and volcanics migrated southward in accord with the oblique collision of the proto-Kuril Arc and the passing of the spreading axis of the Kuril Basin. At the middle-Late Miocene (c), the southward drift of the proto-Kuril Arc ceased, and since then collision tectonics between the Eurasian and the Okhotsk plates has been dominant along the shear zone. This collision produced transpressional shear along the Central Hokkaido–Sakhalin Fault and perhaps accelerated clockwise rotations in the shear zone.

volcanism, from the Early Miocene to the early-Late Miocene (23–9 Ma).

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